# Personality-Aware Engagement Prediction in Online Learning

## Project Report

## Submitted By


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# Abstract

Online learning environments generate large amounts of multimodal behavioral data that can be used to estimate learner engagement. Accurate engagement prediction helps educational platforms improve personalization, intervention strategies, and overall learning outcomes. This project focuses on developing a personality-aware engagement prediction system using multimodal behavioral features extracted from video, audio, and temporal interaction signals.

The project is divided into three major tasks. First, multimodal handcrafted and pretrained features are extracted from behavioral signals using tools such as OpenFace and OpenSMILE. Second, classical machine learning models are explored for engagement prediction using multiple temporal transformation strategies. Third, a personality-conditioned long-range temporal model is proposed to capture temporal dependencies while being robust to noisy labels and missing temporal segments.

The proposed framework combines behavioral dynamics with personality traits to improve engagement prediction across different learners.

Multiple evaluation metrics including Concordance Correlation Coefficient (CCC), F1-score, Mean Absolute Error (MAE), and accuracy are used to assess performance.


# 1.Introduction

The rapid growth of online education platforms has transformed the educational landscape by enabling learners to access educational content remotely. Despite these advantages, maintaining learner engagement remains a major challenge in digital learning environments. Student engagement directly influences attention, participation, retention, and academic performance.

Traditional classroom environments allow instructors to observe learner behavior physically. However, online learning systems lack direct observation capabilities, making automatic engagement prediction an important research area.

Human engagement is multimodal in nature. Facial expressions, eye gaze, head movements, voice patterns, and interaction behaviors collectively provide information about a learner's cognitive and emotional state. Furthermore, personality traits influence how students interact with learning material. Some learners naturally appear expressive and interactive, while others may remain visually passive despite being attentive.

This project aims to design a multimodal machine learning framework capable of predicting learner engagement while incorporating personality information. The proposed system explores both short-term and long-term temporal behavioral modeling.

# 2.Problem Statement

1. **Feature extraction**: Explore multimodal feature representations for engagement prediction. Used OpenFace and OpenSmile
2. **Comparing Classical Model:** Handling noisy labels and missing temporal chunks. Modeling engagement using classical machine learning approaches.
3. **Personality-Conditioned Temporal Modeling:** The primary goal of this project is to develop a personality-aware multimodal engagement prediction framework capable of: Designing a robust temporal model for long-range engagement prediction. Improving prediction generalization across personality groups.

# 3.Dataset Description

The project uses a subset of the MOCSE dataset for engagement prediction.

**Dataset Components**

The dataset contains:

- Video recordings of learners
- Video recordings of learners and Professors (context)
- Engagement annotations
- Personality trait information

# 4.Data Modalities

**Video Modality:** Video frames contain:

- Facial expressions
- Eye gaze direction
- Head pose
- Facial Action Units (AUs)

**Audio Modality:** Audio signals contain:

- Prosodic features
- Pitch variation
- Speech energy
- Spectral characteristics

**Personality Information:** Personality annotations include Introversion, Trust, Emotional Stability, Conscientiousness, Sociability, etc.

# 5.Feature Extraction

## 5.1.OpenFace

OpenFace is a state-of-the-art computer vision toolkit designed for facial behavior analysis. It processes images, video streams, or webcam feeds to comprehensively track and quantify facial movements and expressions over time.

**Features Extracted**

- **Facial Action Units (AUs):** Based on the Facial Action Coding System (FACS), it extracts the presence (0 or 1) and intensity (on a 1-5 scale) of specific muscle movements. For instance, AU04 (Brow Lowerer) and AU12 (Lip Corner Puller) are highly indicative when trying

to classify affective states like confusion, boredom, or engagement.
- **Head Pose:** 3D translation (x, y, z positions) and rotation (pitch, yaw, roll), which helps quantify whether a subject is nodding, looking away, or actively focusing.
- **Eye Gaze:** Direction vectors for both eyes, tracking exactly where a person is looking.
- **Facial Landmarks:** The raw 2D and 3D coordinate data of 68 different points on the face.

## 5.2.OpenSMILE Features

OpenSmile (Speech & Music Interpretation by Large-space Extraction) is a robust audio processing tool that extracts acoustic characteristics from speech. It strips away the semantic meaning of the words (what is being said) and focuses purely on the paralinguistics (how it is being said).

**Features Extracted**

- **Prosodic Features**: Pitch (Fundamental Frequency, F0) and loudness/energy. Monotonous pitch might strongly correlate with a state like boredom, while highly variable pitch might indicate active interest.
- **Spectral Features**: Mel-Frequency Cepstral Coefficients (MFCCs), spectral flux, and spectral centroid. These measure the timbre and "shape" of the sound, capturing differences in vocal tract resonance.
- **Voice Quality:** Metrics like jitter (micro-variations in pitch frequency) and shimmer (micro-variations in vocal amplitude), which can capture nuances like vocal strain or breathiness.

**Since OpenFace and OpenSmile operate at different sampling rates**

**Hence going forward only OpenFace features were used**

# 6.Short-Context Multimodal Modeling

## 6.1.Temporal Representation Strategies

Since behavioral features are temporal sequences, they must be converted into fixed-length representations.

**6.1.1 Temporal Aggregation:** Squash the entire timeline down into a few basic summary statistics.

Instead of feeding the model all the videos, this function looks at the sequence of 10sec numbers and calculates three things:

- Mean
- Variance
- Max.

**6.1.2 Dynamic Feature:** Measure the *rate of change* or volatility.

This function uses `np.diff()`, which subtracts Frame 1 from Frame 2, Frame 2 from Frame 3, and so on. This gives you 49 "deltas" (changes). It then takes the absolute value (so negative dips and positive spikes both count as "movement") and calculates the average of those movements.

**6.1.3 Sliding Window Representation:** Preserve the timeline, but chop it up to give the model more examples of "micro-events."

This is the most complex transformation. Instead of summarizing the all frames, it takes a "window" of 25 frames and slides it across the timeline, moving forward 20 frames at a time. Getting an overlap of 5 frames.

## 6.2.PCA

All 3 datasets with different temporal representation strategies were then made to go through PCA.

PCA works by reorienting your data onto a new set of axes that better capture the "spread" (variance) of the information. Think of it as finding the best possible angle to take a 2D photograph of a 3D object so that you can still recognize what it is.

Internally, PCA is essentially removing **multicollinearity**. If two of your features are 90% correlated (like "Square Footage" and "Number of Rooms" in a house dataset), PCA will likely combine them into a single component, reducing the workload for your machine learning model without losing the underlying "size" signal.

## 6.3 Classical machine learning models

The following ML models were used with different and tested with different parameters. The set of best parameters were used for the comparison.

- Logistic regression
- SVM (Support Vector Machine)

(Parameters each ML model was tested on)

```
param_grids = {
    "Logistic Regression": {
        'C': [0.01, 0.1, 1, 10],
        'solver': ['liblinear', 'lbfgs']
    },
    "SVM": {
        'C': [0.1, 1, 10],
        'gamma': ['scale', 'auto', 0.1],
        'kernel': ['rbf']
    },
    "Random Forest": {
        'n_estimators': [50, 100],
        'max_depth': [None, 10, 20],
        'min_samples_split': [2, 5]
    },
    "XGBoost": {
        'learning_rate': [0.01, 0.1],
        'max_depth': [3, 6],
        'n_estimators': [100, 200]
    },
    "MLP": {
        'hidden_layer_sizes': [(64, 32), (128, 64)],
        'learning_rate_init': [0.001, 0.01]
    }
}
```

- Random Forest
- XGBoost
- MLP (Neural Network)

| Evaluation Method | Model | Best Parameters | Train Accuracy | Train F1-Score | Test Accuracy | Test F1-Score |
|---|---|---|---|---|---|---|
| **Temporal Aggregation PCA** | Logistic Regression | C: 0.1, solver: 'lbfgs' | 0.5276 | 0.5246 | 0.4958 | 0.4957 |
| | SVM | C: 10, gamma: 'auto', kernel: 'rbf' | 0.9851 | 0.9851 | 0.4873 | 0.4794 |
| | Random Forest | max_depth: 10, min_samples_split: 5, n_estimators: 100 | 0.9735 | 0.9734 | 0.4873 | 0.4868 |
| | XGBoost | learning_rate: 0.01, max_depth: 3, n_estimators: 100 | 0.6518 | 0.6516 | 0.5042 | 0.5027 |
| | MLP | hidden_layer_sizes: (128, 64), learning_rate_init: 0.001 | 0.9745 | 0.9745 | 0.5 | 0.4987 |
| **Dynamic Features PCA** | Logistic Regression | C: 0.01, solver: 'liblinear' | 0.5382 | 0.5279 | 0.4534 | 0.438 |
| | SVM | C: 1, gamma: 0.1, kernel: 'rbf' | 0.6306 | 0.6257 | 0.4492 | 0.4388 |
| | Random Forest | max_depth: 10, min_samples_split: 2, n_estimators: 100 | 0.9756 | 0.9756 | 0.4576 | 0.4576 |
| | XGBoost | learning_rate: 0.01, max_depth: 3, n_estimators: 100 | 0.6582 | 0.6582 | 0.4449 | 0.4432 |
| | MLP | hidden_layer_sizes: (128, 64), learning_rate_init: 0.01 | 0.6794 | 0.6691 | 0.5085 | 0.4981 |
| **Sliding Window PCA** | Logistic Regression | C: 10, solver: 'liblinear' | 0.5267 | 0.5266 | 0.5348 | 0.5348 |
| | SVM | C: 10, gamma: 'auto', kernel: 'rbf' | 0.9817 | 0.9817 | 0.8145 | 0.8124 |
| | Random Forest | max_depth: None, min_samples_split: 2, n_estimators: 100 | 0.983 | 0.983 | 0.7992 | 0.7991 |
| | XGBoost | learning_rate: 0.1, max_depth: 6, n_estimators: 200 | 0.9308 | 0.9308 | 0.7424 | 0.7423 |
| | MLP | hidden_layer_sizes: (128, 64), learning_rate_init: 0.001 | 0.8989 | 0.8986 | 0.7525 | 0.7514 |

# 7.Personality-Conditioned Temporal Modeling

## 7.1 Objective

The goal of this task is to design a long-range temporal model capable of robust engagement prediction while accounting for personality traits, noisy labels, and missing temporal segments.

## 7.2 Proposed Temporal Architecture

A Personality aware Bi-directional LSTM trained with huber loss to make it resistant to noisy labels was used.

### Input Components

- Multimodal behavioral features
- Personality trait vectors

### Model Components

1. A linear layer to transform the professor and student personality components.
2. A Bi-directional LSTM with two hidden layers with 128 neurons and a 30 percent dropout rate.
3. A fully connected feed forward network to turn the LSTM output into a final prediction between 0 and 1.

## 7.3 Personality Conditioning

Personality embeddings were integrated with behavioral features.

Benefits:

- Better personalization
- Improved behavioral interpretation
- Enhanced generalization

Example:

- Extroverted learners may show stronger visible engagement.
- Introverted learners may appear visually passive despite active learning.

## 7.4 Handling Noisy Labels

### Robust Loss Functions

We used Huber loss to make sure that the impact of noisy labels was minimal.

## 7.5 Handling Missing Temporal Segments

Missing chunks were addressed using Dynamic Padding. We added dynamic padding in our data loading functions to ensure that the model gets the number of expected features.

# 8. Results and Analysis

## 8.1 Classical ML Model Performance

### Observation

We see that most Temporal techniques give accuracy close to 50%, which is equivalent to random guessing. The best temporal representation technique (which will be revealed later, stay tuned) gave an accuracy between 74 and 80 percent, except in the case of logistic regression. SVM preformed the best out of all of them, with an accuracy of around 81 percent.

## 8.2 Temporal Representation Comparison

### Observation

Sliding-window was by far the best strategy out of the three used, being the only one that allowed for accuracy significantly more than random guessing.

### 8.3 Personality-Aware Temporal Model Results

#### Observation

Giving personality information and using a larger sequential model (LSTM) gave an accuracy of around 60 percent, not being able to outclass SVMs with sliding window representation.

## 9. Challenges Faced

Several challenges were encountered during the project:

### 9.1 Data Imbalance

Some engagement labels were underrepresented, so a balanced subset had to be created.

### 9.2 Missing Temporal Segments

Some recordings had incomplete sequences so we needed to pad them out with zeros.

### 9.3 Annotation Noise

Human engagement annotations were subjective. Thus, robust loss functions were used to combat noise.

### 9.4 High Dimensional Features

Multimodal features increased computational complexity, so we used pca to select for features

## 10. Conclusion

This project developed a personality-aware multimodal framework for predicting online learner engagement. While the deep learning Bi-LSTM successfully integrated static psychological traits and long-range behaviors, the classical SVM with sliding windows ultimately achieved higher accuracy (80% vs. 60%), demonstrating that localized, short-term behavioral signals were more reliable predictors for this specific dataset than long-range temporal modeling.

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